

Slow-light and high-sensitivity terahertz sensing enabled by multifunctional resonant metasurfaces

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Resonant metasurfaces provide a promising platform for manipulating light–matter interactions at terahertz frequency, enabling functionalities such as slow-light propagation and high-sensitivity sensing. Here, we demonstrate a symmetry-controlled metasurface that supports a tunable transition between quasi-bound states in the continuum (quasi-BICs) and an electromagnetically induced transparency (EIT)-like response. By varying the rotation angle of the meta-atoms, the system evolves from an isolated quasi-BIC state to a strongly coupled regime where the quasi-BIC coherently interacts with an electric dipole mode, giving rise to a pronounced EIT window. Multipole decomposition reveals the evolution of the radiative channels and clarifies the mechanism underlying the BIC–EIT transition. Owing to the hybrid spectral response, the metasurface simultaneously supports slow-light propagation with group delays up to 30.3 ps in simulations and 14.2 ps experimentally, and high-sensitivity terahertz sensing enabled by the quasi-BIC. As a proof-of-concept application, *in situ* detection of furfural in transformer insulating oil is demonstrated with concentrations down to 1 ppm. These results establish symmetry-controlled metasurfaces as a versatile platform for integrating coherent wave manipulation and ultrasensitive sensing in terahertz photonic devices. © 2026 Chinese Laser Press

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1. INTRODUCTION

Resonant metasurfaces, composed of subwavelength artificial meta-atoms engineered to tailor electromagnetic responses, have rapidly emerged as powerful platforms for light manipulation [1–5]. Owing to their structural compactness and high design degree freedom, metasurfaces have enabled transformative advances in optical imaging [6,7], sensing [8–10], on-chip communication [11], and nonlinear optics [12]. Recent progress has further propelled the field toward multifunctional and highly integrated photonic chips capable of simultaneous signal modulation and information processing [13,14]. Realizing such advanced functionalities has motivated extensive efforts to emulate quantum-inspired phenomena within classical photonic architectures, including Fano resonances [15], spin–orbit interactions [16,17], electromagnetically induced transparency (EIT) [18–20], and bound states in the continuum (BICs) [21–23].

Precise control of resonant spectral responses in metasurfaces plays a key role in enhancing light–matter interactions and enabling advanced terahertz photonic functionalities. Particularly, slow-light manipulation and high-sensitivity sens-

ing represent two important capabilities for information processing and molecular detection. Slow-light effects rely on sharp spectral dispersion that dramatically reduces the group velocity of electromagnetic waves [24], while ultrasensitive sensing requires resonances with narrow linewidths and strong field confinement to amplify the response to environmental perturbations [25]. Among various approaches to spectral engineering, interference-induced transparency and high-*Q* resonances have attracted particular attention [26]. Interference between radiative modes can produce sharp transparency windows accompanied by steep dispersion, enabling slow-light effects [27]. Meanwhile, BICs provide an effective route to achieving ultra-high-*Q* resonances through suppressed radiation loss, which is highly desirable for sensing applications. Bridging these distinct resonant regimes through controllable modal interactions, thus, provides a promising route toward metasurface platforms that integrate slow-light propagation with high-sensitivity detection. Achieving these functionalities within a single metasurface remains challenging, as they generally rely on different resonance mechanisms, spectral responses, and field-coupling conditions, which are often difficult to simultaneously optimize within a single metasurface. Consequently, engineering metasurfaces that

can flexibly tailor resonant profiles and modal interactions has become an important strategy for integrating multiple photonic functionalities.

Here, we demonstrate a symmetry-engineered metasurface that enables a continuous transition between quasi-BIC and EIT-like response. By introducing rotational symmetry breaking in the meta-atoms, the initially non-radiative quasi-BIC mode becomes radiatively accessible and coherently couples with a broadband electric dipole (ED) resonance. The resulting modal interference produces a pronounced transparency window with steep spectral dispersion, leading to a significant slow-light effect, with maximum group delays of 30.3 ps and 14.2 ps in simulations and experiments, respectively. Beyond slow-light manipulation, the proposed platform also enables practical terahertz sensing in complex environments. We experimentally demonstrate direct and rapid detection of furfural concentration in transformer insulating oil, achieving a detection resolution down to 1 ppm. The proposed metasurface provides a unified platform for integrating slow-light manipulation and terahertz sensing, offering new opportunities for multifunctional terahertz photonic devices.

2. RESULTS AND DISCUSSION

Figure 1 summarizes the design, fabrication, and characterization of the proposed resonant membrane metasurface. The three-dimensional schematic shown in Fig. 1(a) depicts a periodic array of double copper rectangular resonators patterned onto a 50- μm -thick free-standing cyclic olefin copolymer (COC) membrane film, which serves as a low-loss and mechanically flexible substrate with a permittivity of $\epsilon = 2.32$. The thickness of the copper layer is 200 nm. The detailed geometric parameters of the unit cell are depicted in Fig. 1(b). The meta-atoms are periodically arranged along x and y directions with lattice constants of $P = 300\ \mu\text{m}$. The rectangular resonators possess a width of $a = 80\ \mu\text{m}$ and a length of $b = 100\ \mu\text{m}$.

To realize a controllable transition between quasi-BIC and EIT-like response, we introduce a spectral engineering regime via tailoring the radiative leakage through in-plane symmetry breaking (rotation angle β). This interaction can be described by a three-level coupling model, which is illustrated in Fig. 1(d). In the symmetric configuration, the two rectangular resonators form a dark resonance with forbidden radiation, giving rise to a high- Q BIC resonance, which induces strong local electromagnetic field confinement at the resonance frequency [Fig. 1(e)]. Such intense field localization significantly enhances light-matter interactions, thereby establishing a robust platform for high-sensitivity sensing applications. When the right resonator is gradually rotated while the left one remains fixed, the in-plane symmetry of the unit cell is perturbed, opening radiative channels that enable controlled coupling between the originally dark resonance and a broadband bright ED mode. As the asymmetry increases, the system evolves from an isolated quasi-BIC toward a strongly interacting regime where mode interference dominates the spectral response. The resulting destructive interference not only gives rise to a distinct transparency window [Fig. 1(f)] but also to a slow-light effect.

To demonstrate the spectral response and resonance characteristics, numerical simulations were performed using a finite-element numerical calculation method. Periodic Floquet boundary conditions were applied in the x and y directions, while open boundaries were set along the z direction to emulate free-space propagation. Adaptive tetrahedral mesh refinement was employed to ensure numerical accuracy and convergence, particularly in regions with strong field gradients. The simulated spectral response and corresponding electromagnetic field distributions of the proposed metasurface were illustrated in Figs. 2(a) and 2(b). An x -polarized plane wave was incident along the z direction, with the electric field oriented parallel to the short axis of the rectangular resonators. Figure 2(a) depicts the electric field distributions and surface current profiles of the two representative modes at a specific rotation angle of

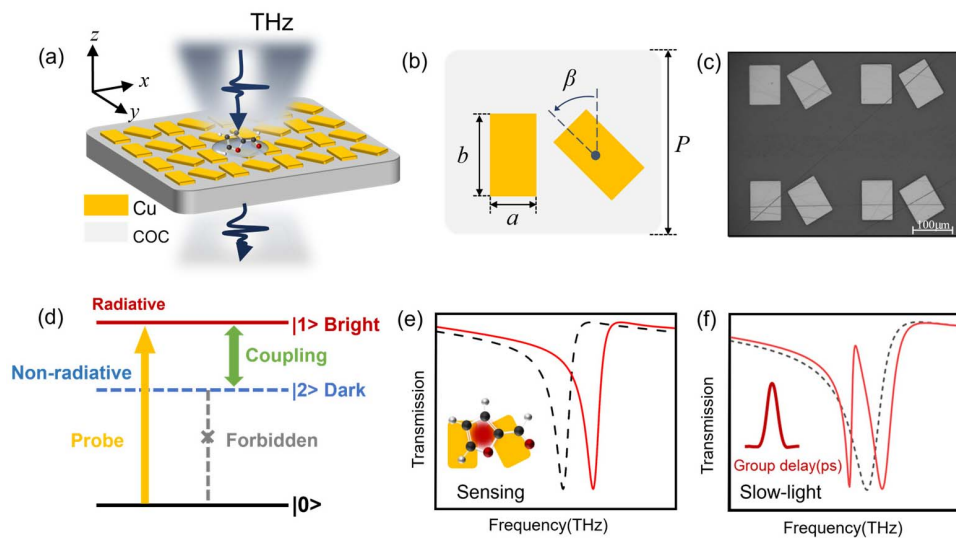


Fig. 1. Design, fabrication, and characterization of the terahertz metasurface. (a) Three-dimensional (3D) schematic of the proposed metasurface. (b) Geometry of the unit cell with detailed dimensions. (c) Optical microscope image of the fabricated metasurface. (d) Schematic diagram of the three-level atomic system model. (e) Illustration of the sensing mechanism utilizing local field enhancement. (f) Transmission spectrum exhibiting a sharp transparency window induced by the quasi-BIC, accompanied by the slow-light effect.

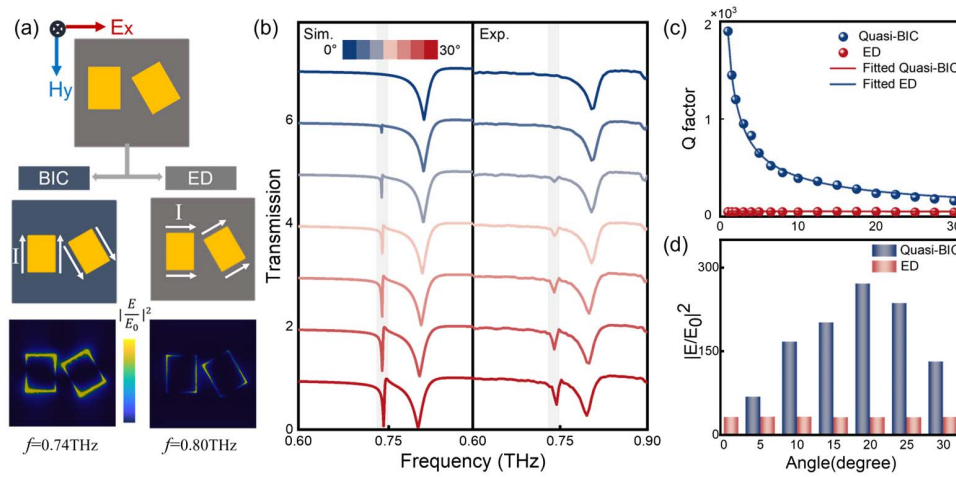


Fig. 2. Electromagnetic properties and spectral evolution of the terahertz metasurface. (a) Near-field characteristics at $\beta = 30^\circ$. The upper panels illustrate the surface current distributions for the quasi-BIC and ED modes; the lower panels display the corresponding normalized electric field distributions. (b) Comparison between the simulated (left) and experimental (right) transmission spectra as the rotation angle β varies from 0° to 30° . (c) Evolution of the Q -factors for both modes as a function of the rotation angle β . The solid lines represent the fitting results. (d) Comparison of the maximum electric field enhancement factors for the quasi-BIC and ED modes at different rotation angles.

$\beta = 30^\circ$. For resonance I at 0.74 THz, anti-parallel surface currents on the paired resonators produce a circulating current loop characteristic of magnetic dipole (MD) oscillation. Owing to rotational symmetry breaking, the originally symmetry-protected BIC is transformed into a radiative quasi-BIC that supports strong electromagnetic confinement with suppressed radiation leakage, thereby enhancing light-matter interactions with trace analytes, while for resonance II at 0.80 THz, surface currents flow in parallel on the two resonators and are aligned with the incident electric field, forming a typical ED oscillation that can be directly excited by the incident wave.

To characterize the evolution of the quasi-BIC under symmetry breaking, the transmission spectra were simulated as a function of the rotation angle β ranging from 0° to 30° , as shown in Fig. 2(b). In the symmetric configuration ($\beta = 0^\circ$), only a broad resonance dip appears around 0.80 THz corresponding to the radiative ED, while the transmission near 0.74 THz approaches unity, which indicates the presence of a symmetry-protected BIC that remains completely decoupled from the radiation continuum. As β gradually increases, the rotational perturbation breaks the in-plane symmetry and opens radiative channels for the previously dark BIC mode. Consequently, a sharp resonance dip emerges near 0.74 THz, marking the formation of a quasi-BIC. As β gradually increases, the radiative coupling strength between the quasi-BIC and free space becomes progressively stronger, resulting in gradual linewidth broadening and enhanced resonance modulation. Notably, compared with many previously reported designs that exhibit strong sensitivity to geometric perturbations [28–30], the proposed metasurface demonstrates remarkable spectral robustness. The quasi-BIC undergoes an almost negligible frequency shift of only 0.0026 THz across the investigated rotation range. To experimentally display the performance of the proposed membrane metasurface, we fabricated the samples using standard photolithography, and

the optical image of the samples can be seen in Fig. 1(c). The corresponding experimental transmission spectra measured using a terahertz time-domain spectroscopy (THz-TDS) system are shown in the right panel of Fig. 2(b). The measured results exhibit excellent agreement with the simulations, reproducing the resonance frequencies, spectral evolution, and linewidth variation of resonances I and II. A slight linewidth broadening at quasi-BIC resonance is observed and experimentally compared with the simulated results, which could be attributed to additional scattering losses induced by inevitable fabrication imperfections, as well as the finite spectral resolution of the THz-TDS system.

To quantitatively analyze the resonance evolution, the transmission spectra were fitted using the classical Fano formalism [31],

$$T(\omega) = T_0 + A_0 \frac{[q + 2(\omega - \omega_0)/\Gamma]^2}{1 + [2(\omega - \omega_0)/\Gamma]^2},$$

where ω_0 is the resonant frequency, Γ is the resonance linewidth, T_0 is the transmission offset, A_0 is the continuum-discrete coupling constant, and q is the Breit-Wigner-Fano parameter determining asymmetry of the resonance profile. The excellent agreement between the full-wave simulations and the nonlinear Fano fitting confirms the characteristic scaling behavior associated with the transition from an ideal BIC to a quasi-BIC, as presented in Fig. 2(c). From the extracted resonance parameters, the quality factor exhibits a typical angular dependence of symmetry-protected BICs, which decreases from a peak value of 1.9×10^3 to 1.6×10^2 , while resonance II maintains a relatively low and nearly invariant Q -factor owing to strong radiative damping. Correspondingly, the normalized field enhancement factors $|E/E_0|^2$ (where E_0 is the incident field amplitude) for both modes were evaluated, as shown in Fig. 2(d). The quasi-BIC exhibits substantially stronger field enhancement than resonance II. Interestingly, the field enhancement reaches a maximum around $\beta \approx 20^\circ$

rather than increasing monotonically as β increases, due to the trade-off between intrinsic loss and radiation loss.

Building upon the above analysis, the proposed metasurface simultaneously supports a broadband ED resonance and a high- Q quasi-BIC with markedly different radiative properties. The interaction between these two modes can be effectively tuned by varying the rotation angle β . Figure 3(a) presents the evolution of the resonance frequencies as a function of β , where the angular range can be divided into two regimes: an independent region and a coupled region. In the independent region [left side of the dashed line in Fig. 3(a)], the ED mode (red curve) exhibits a pronounced redshift with increasing β , while the quasi-BIC (blue curve) remains nearly spectrally invariant. The large frequency detuning between the two resonances suppresses modal interaction, and the system, therefore, behaves as two essentially independent oscillators. As β further increases, the spectral detuning gradually decreases, and the system approaches a near-resonant condition. This transition marks the onset of the coupled regime [right side of the dashed line in Fig. 3(a)], where strong modal hybridization and coherent interaction can occur [32].

To further clarify the physical origin of this coupling process, we performed the normalized multipole decomposition to analyze the behavior of resonances I and II, which is displayed in Fig. 3(b). The quasi-BIC [left panel of Fig. 3(b)] exhibits a clear multipolar evolution. Within the independent region, the resonance is primarily governed by MD and electric-quadrupole (EQ) contributions, while the ED component is strongly suppressed. This multipolar configuration corresponds to a typical “dark” mode that is weakly coupled to the radiation continuum. For the quasi-BIC, the ED contribution is nearly negligible at $\beta = 0^\circ$ and increases continuously to approximately 95% at $\beta = 90^\circ$. Conversely, as β increases from 5° to 90° , the contributions from the MD and EQ drop sharply from 52% to 2% and from 40% to 2%, respectively. These results provide a quantitative demonstration of the activation of the radiative ED channel as symmetry is broken. In contrast, the optical response of resonance II [right panel of Fig. 3(b)] remains completely dominated by the ED contribution across the entire angular range, confirming its intrinsically radiative “bright” nature. With the increment of β , the arising of a new ED mode at resonance I leads to an increasing spatial

and radiative overlap with the ED mode at resonance II. When this activated radiative ED channel spectrally overlaps with the broadband ED resonance (resonance II), both modes share the same radiation channel but carry opposite phases. Their destructive interference suppresses radiation at the resonance frequency, ultimately giving rise to the EIT-like transparency window observed in the transmission spectrum. This underlying strong coupling regime has been rigorously verified by the analytical model provided in Appendix A.

The coherent coupling between the broadband ED and the quasi-BIC becomes evident in the transmission spectra as the rotation angle β increases from 40° to 90° . As shown in Fig. 4(a), a pronounced EIT-like transparency window emerges at both simulation and experiment spectra once the system enters the coupled regime where spectral proximity and modal overlap are established. The underlying mechanism can be understood using the theory of coupled radiative channels together with an analogy to a three-level atomic system [Fig. 1(d)] [33]. Resonance II functions as a “bright” oscillator directly excited by incident light, analogous to the $|0\rangle \rightarrow |1\rangle$ transition in an atomic system. Conversely, resonance I acts as a “dark” oscillator that does not couple directly to free space, analogous to the forbidden $|0\rangle \rightarrow |2\rangle$ transition. With the introduction of asymmetry, near-field interactions enable energy transfer from the bright mode to the quasi-BIC, thereby opening an intermediate excitation pathway: $|0\rangle \rightarrow |1\rangle \rightarrow |2\rangle \rightarrow |1\rangle$. The coexistence of the direct excitation channel and this intermediate coupling pathway leads to destructive interference at a specific frequency ($f \approx 0.754$ THz), which suppresses the radiative loss of the bright mode. As a result, the scattering from the metasurface is strongly reduced, producing a sharp transmission peak that constitutes the EIT-like transparency window.

To investigate the mechanism of the destructive interference, a representative case with $\beta = 60^\circ$ was selected, as illustrated in Fig. 4(b). The transmission spectrum exhibits a typical Fano-type EIT profile, featuring a sharp transparency peak (red circle) located between two resonance dips (red square and red diamond). The electric field distributions at these three characteristic frequencies are presented in Fig. 4(c). At both the low- and high-frequency resonance dips, strong local electric field enhancement is observed within the metasurface. The hybridized resonant modes with significant radiative leakage

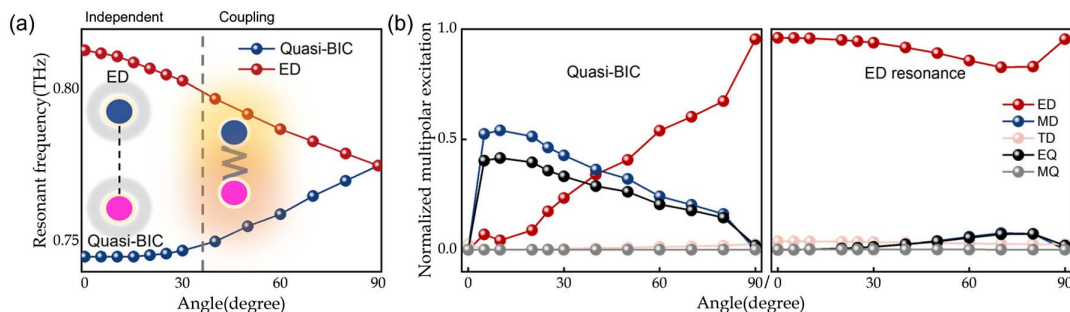


Fig. 3. Evolution of resonance characteristics and multipole decomposition. (a) Resonant frequencies of the quasi-BIC (blue spheres) and the ED (red spheres) as a function of the rotation angle. The regions on either side of the dashed line denote the independent and coupled regimes, respectively. The inset schematically illustrates the corresponding coupled oscillator model. (b) Multipole decomposition of the scattered power for the quasi-BIC (left panel) and the ED resonance (right panel). The curves represent the contributions from the ED, MD, toroidal dipole (TD), EQ, and magnetic quadrupole (MQ), respectively.

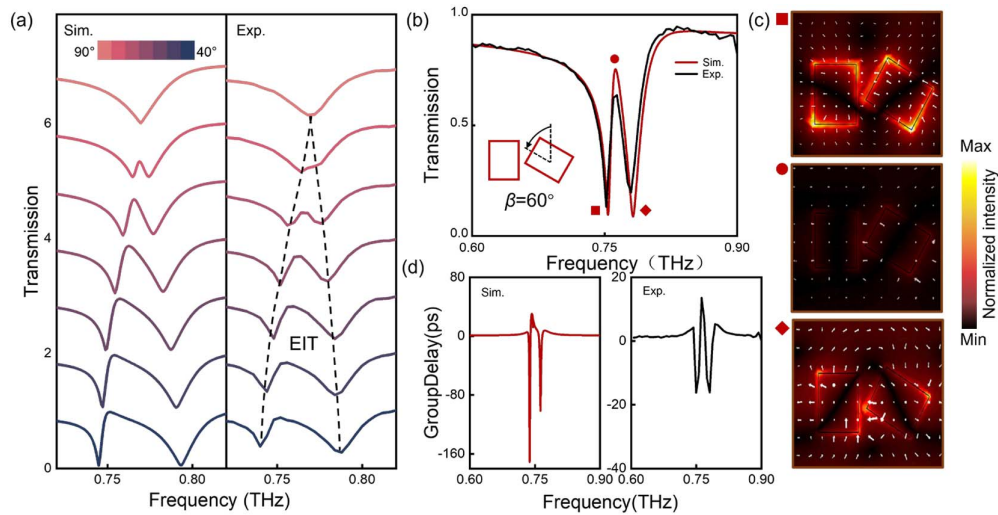


Fig. 4. EIT-like response and slow-light performance. (a) Evolution of the simulated and experimentally measured transmission spectra within the strong coupling regime. (b) Transmission spectra at $\beta = 60^\circ$, exhibiting a typical EIT profile. (c) Normalized electric field distributions at the three key frequencies marked in (b). (d) Simulated (left panel) and experimentally measured (right panel) group delay spectra.

at these two dips suppress the transmission of resonances. In contrast, at the transparency peak (red circle), the near-field excitation around the resonators is strongly reduced, which reflects the destructive interference between the bright ED and the quasi-BIC. This interference suppresses the radiative scattering of the metasurface and allows the incident wave to pass through the structure with minimal attenuation, thereby producing the high-transmission EIT window.

The transparency window is associated with steep normal dispersion, which typically gives rise to a slow-light response in EIT systems [34]. To quantitatively evaluate this phenomenon, the group delay τ_g was calculated from the frequency derivative of the transmission phase [35],

$$\tau_g = -d\phi/d\omega,$$

where ϕ is the transmission phase and ω is the angular frequency. Experimentally, the group delay was derived from THz-TDS time-domain signals obtained with a 400 ps scan length, providing a high spectral resolution of approximately 2.5 GHz to capture the rapid phase variations. Figure 4(d) displays the simulated and measured group delay spectra of the metasurface. A pronounced delay peak appears at the center of the transparency window, indicating a significant reduction of the group velocity of the transmitted terahertz wave. At the resonance frequency, group delays of 30.3 ps in simulation and 14.2 ps in experiment are obtained. The smaller experimental value is mainly attributed to linewidth broadening, which diminishes the steepness of the phase derivative.

To further quantify the slow-light performance, the delay-bandwidth product (DBP) was evaluated. The experimental transparency window exhibits a bandwidth of 0.0154 THz, yielding a competitive DBP of 0.219. Meanwhile, the giant reduction of group velocity translates to a high experimental group index of $n_g = c/H\tau_g = 85$ based on the propagation length H [36]. Furthermore, a slow-light figure of merit was introduced as $\text{FOM}_{\text{slow}} = \tau_g \times T(\omega_0)$ [37], which simultaneously evaluates

the achievable group delay and transmission efficiency. Owing to the reduced radiative loss associated with quasi-BIC resonance, a relatively high experimental transmission of 0.76 is preserved at the transmission peak, yielding a competitive FOM_{slow} of 10.8 ps. Nevertheless, the experimentally observed group delay clearly demonstrates that the quasi-BIC-mediated EIT metasurface enables temporal control of terahertz wave propagation, which is promising for slow-light devices, optical buffering, and dynamic signal processing.

3. APPLICATION

To evaluate the sensing capability of the proposed metasurface, the detection of furfural concentration in electrical insulating oil was selected as a proof-of-concept application. Furfural is widely recognized as a critical by-product of cellulose thermal degradation in oil-immersed power equipment. Its concentration serves as a quantitative indicator of insulation aging and operational reliability [38,39]. Currently, the industrial detection of dissolved furfural in transformer oil predominantly relies on high-performance liquid chromatography (HPLC), ultraviolet-visible (UV-Vis) spectrophotometry, and colorimetric analysis [40–43]. However, these techniques typically require laborious sample pre-treatment procedures and the consumption of costly chemical reagents, thereby not only increasing operational complexity but also introducing an inherent risk of chemical contamination. In contrast, terahertz metasurface technology has emerged as a label-free and non-destructive alternative that bypasses the need for complex pre-treatment, offering a promising solution to these long-standing challenges. To achieve *in situ* liquid-phase detection, a customized THz-TDS system was constructed, as schematically illustrated in Fig. 5. A metasurface with a rotation angle of 45° was employed as the sensing platform. Pure furfural (ACS $\geq 99\%$) was dissolved in standard transformer oil to prepare gradient solutions ranging from 0 to 50 ppm. In the experiment, 10 μL of the solution was dispensed onto the metasurface and covered with an ultra-thin polyimide film. This film forms a

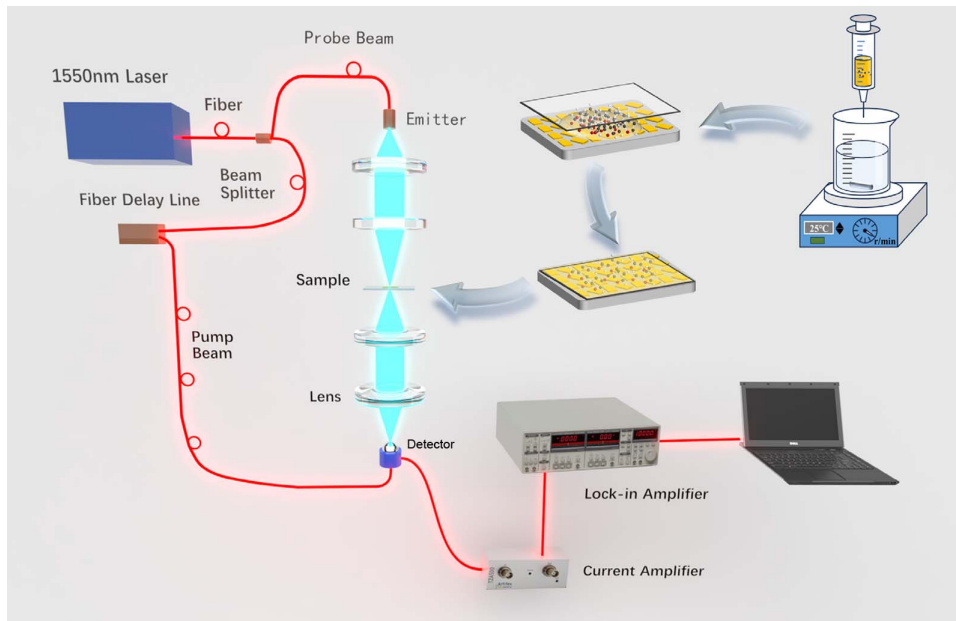


Fig. 5. Schematic of the optical measurement setup and the fabrication flow of test samples.

uniform microfluidic layer that minimizes optical path length variations while simultaneously preventing external contamination. Each reported data point represents the average of 10 independent measurements.

Figure 6(a) depicts the evolution of the terahertz transmission spectra as the furfural concentration increases from 0 to 50 ppm. A clear redshift of both the quasi-BIC and ED reso-

nances is observed with increasing concentration. This spectral shift is attributed to the introduction of furfural molecules into the sensing environment, which increases the effective refractive index of the surrounding medium and consequently perturbs the local dielectric environment of the metasurface. Figure 6(b) summarizes the concentration-dependent frequency shifts (Δf), providing a quantitative measure of the

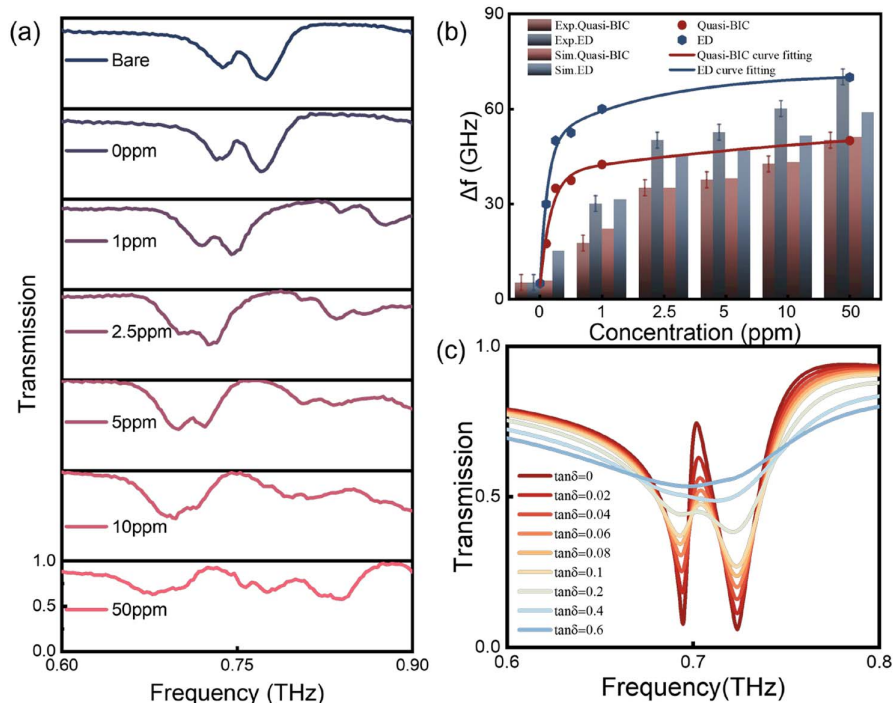


Fig. 6. Sensing spectral response and material parameter analysis. (a) Evolution of the transmission spectra with increasing furfural concentration (0–50 ppm). (b) Statistical analysis of the frequency shifts for the quasi-BIC and ED modes relative to the pure oil baseline. (c) Influence of varying $\tan \delta$ on the transmission spectra.

Table 1. Fitted Parameters of the Modified Langmuir Adsorption Model for Different Resonance Modes

Resonance Mode	$\Delta f_{\max}$ (GHz)	K_L (ppm ⁻¹)	y_0 (GHz)	R^2
ED	65.23	0.68	4.94	0.9884
Quasi-BIC	46.94	0.52	4.38	0.9801

sensing sensitivity. The experimental measurements show excellent agreement with the simulated results for both the quasi-BIC (red bars) and ED (blue bars) modes. Notably, even at an ultra-low concentration of 1 ppm, a clearly resolvable frequency shift can be detected, highlighting the high sensitivity of the platform. To rigorously evaluate this performance, the limit of detection (LOD) was quantitatively defined on the basis of the experimental noise floor. The resonance frequency of pure insulating oil was monitored over 10 independent measurements to determine the spectral noise level (σ). According to the conventional 3σ criterion ($\text{LOD} = 3\sigma/S$), where S is the sensitivity at low concentrations [44], the LOD of the proposed sensor was estimated to be approximately 0.25 ppm. Moreover, a distinct step-like variation between 1 ppm and 2.5 ppm is observed. As the concentration further increases, the frequency shift gradually approaches saturation and deviates from a linear response. To elucidate the origin of this nonlinear behavior, the experimental frequency shifts were fitted using a modified Langmuir adsorption isotherm model [45],

$$\Delta f = \frac{\Delta f_{\max} \cdot K_L \cdot C}{1 + K_L \cdot C} + y_0,$$

where $\Delta f_{\max}$ denotes the maximum frequency shift at full monolayer saturation, K_L is the Langmuir adsorption equilibrium constant characterizing molecular affinity, and y_0 denotes the intrinsic baseline frequency offset at zero concentration. The extracted fitting parameters are summarized in Table 1. The high coefficients of determination ($R^2 = 0.9884$ for the ED mode and 0.9801 for the quasi-BIC mode) indicate that the sensing mechanism is predominantly governed by localized surface adsorption of furfural molecules, rather than by bulk refractive-index modulation. Although the quasi-BIC mode exhibits a comparatively smaller absolute frequency shift, its ultrahigh quality factor and sharply defined spectral response enable a superior tracking precision and a lower detection limit.

In addition to frequency shifts, significant variations in spectral modulation depth are also observed in Fig. 6(a). In particular, the EIT window becomes significantly weakened and nearly vanishes at 50 ppm. The underlying mechanism is further elucidated by numerical simulations of the transmission spectra with varying dielectric loss tangents ($\tan \delta$), as shown in Fig. 6(c). As $\tan \delta$ increases from 0 to 0.6, the EIT peak is strongly suppressed and ultimately disappears, accompanied by pronounced linewidth broadening. This trend closely reproduces the experimental observations, confirming that high-concentration furfural solutions introduce additional dielectric loss in conjunction with refractive-index modification, thereby increasing the damping of quasi-BIC resonances. Owing to its pronounced spectral response and a calculated LOD, the proposed metasurface enables the precise resolution of trace-level furfural concentrations within the 1–5 ppm regime. Such sen-

sitivity is particularly significant for the early diagnosis of insulation aging in transformer systems. As summarized in Table 2, a comparison with representative terahertz metasurface sensors indicates that the proposed platform achieves a well-balanced performance across multiple critical metrics, including the sensitivity, detection limit, and spectral selectivity.

4. CONCLUSION

In summary, we have proposed and experimentally demonstrated a terahertz metasurface that enables a controllable transition from symmetry-protected BICs to an EIT-like response through rotational symmetry breaking. This transition is governed by the activation of a radiative quasi-BIC channel and its coherent interaction with broadband ED resonance, as described by a coupled-mode theory model. The resulting interference produces a sharp transparency window with strong phase dispersion, enabling slow-light behavior with group delays of 30.3 ps in simulations and 14.2 ps in experiments. In addition, the platform enables *in situ* detection of trace furfural in transformer oil with an LOD down to 0.25 ppm. Beyond these specific demonstrations, the key implication of this work is a general design strategy for integrating symmetry-engineered quasi-BICs with broadband resonances to achieve simultaneous control of resonance line-shape, dispersion, and sensing functionality within a single metasurface platform. The presented approach provides a promising route toward multifunctional terahertz metasurfaces for wave manipulation and industrial sensing applications.

APPENDIX A: COUPLED OSCILLATOR MODEL FOR ELUCIDATING STRONG COUPLING

Within the metasurface unit cell, the resonance frequencies of the supported quasi-BIC and ED modes are primarily governed by the lattice periods along the x and y axes, respectively. To further explore the dependence of coupling strength on the structural asymmetry, the transmission spectra for various rotation angles (β ranging from 0° to 45°) are presented in Fig. 7. In the symmetric configuration ($\beta = 0^\circ$), the quasi-BIC is inhibited due to symmetry protection, leaving only the ED, which exhibits a characteristic redshift with increasing P_y . When $\beta \neq 0^\circ$, the symmetry breaking activates the quasi-BIC channel, enabling its radiative coupling. Consequently, a pronounced anticrossing behavior (Rabi splitting) emerges between the ED and quasi-BIC, arising from their strong mutual interaction. Notably, the Rabi splitting energy increases monotonically with the rotation angle β , indicating an enhancement in coupling strength. This phenomenon can be quantitatively elucidated by the coupled oscillator model, where the eigenvalue equation of the coupled system is expressed as [32]

$$\begin{pmatrix} E_{\text{ED}} - i\gamma_{\text{ED}} & g \\ g & E_{\text{qBIC}} - i\gamma_{\text{qBIC}} \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = E_{\pm} \begin{pmatrix} u \\ v \end{pmatrix},$$

where E_{ED} and E_{qBIC} represent the resonance energies of the uncoupled ED and the quasi-BIC, respectively. g denotes the coupling strength of two modes. γ_{ED} and γ_{qBIC} are their respective half-bandwidths. $E_{\pm}$ stands for the eigen energies of two new hybrid modes. The eigenvalue equation allows us to obtain the eigen energies of two new hybrid states as follows:

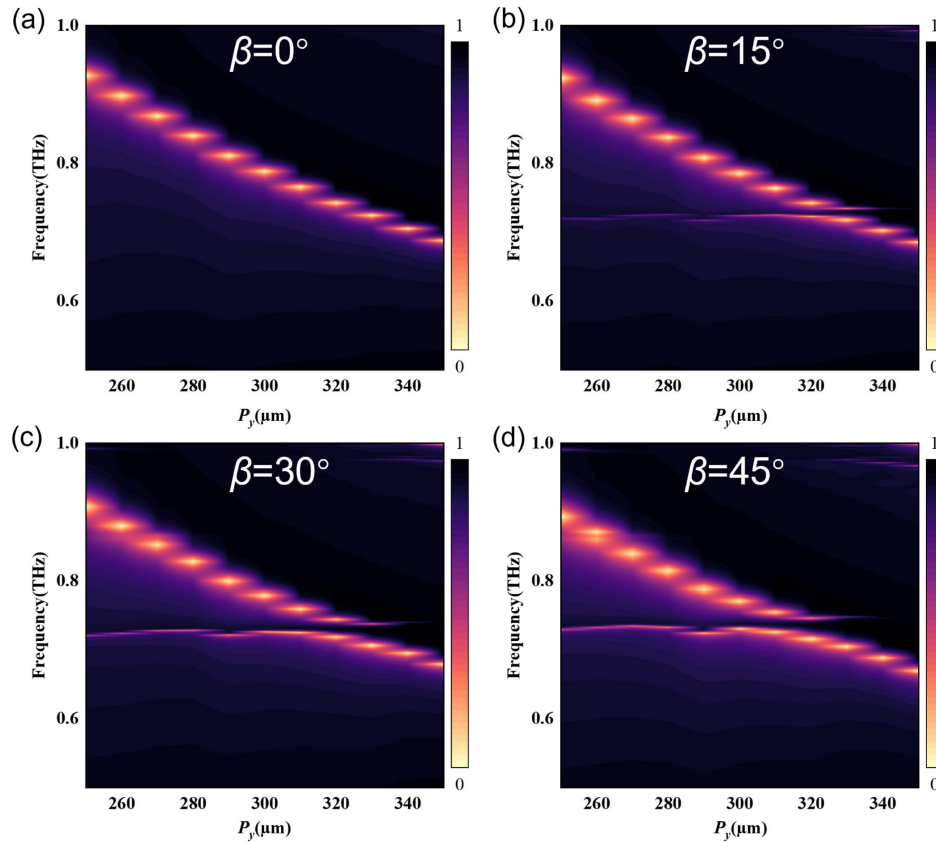


Fig. 7. Transmission spectra of the hybrid system as a function of the rotation angle. (a)–(d) correspond to β values from 0° to 45° with an increment of 15° .

$$E_{\pm} = \frac{(E_{\text{ED}} - i\gamma_{\text{ED}}) + (E_{\text{qBIC}} - i\gamma_{\text{qBIC}})}{2} \pm \sqrt{g^2 + \frac{1}{4}[(E_{\text{ED}} - i\gamma_{\text{ED}}) - (E_{\text{qBIC}} - i\gamma_{\text{qBIC}})]^2}.$$

Rabi splitting ($\hbar\Omega$), which represents the minimum energy separation between the two hybrid branches at zero detuning ($E_{\text{ED}} - E_{\text{qBIC}} = 0$), is related to the coupling strength g by

$$\hbar\Omega = 2g.$$

According to the criterion for strong coupling, the interaction must satisfy the following condition:

$$\hbar\Omega > \frac{\hbar\gamma_{\text{qBIC}} + \hbar\gamma_{\text{ED}}}{2}.$$

Taking $\beta = 45^\circ$ as a representative example, the Rabi splitting energy extracted from the transmission spectra in Fig. 7(d) is $\hbar\Omega = 79.4 \mu\text{eV}$. For the two uncoupled resonances, the damping rates are determined to be $\hbar\gamma_{\text{qBIC}} = 32.5 \mu\text{eV}$ and $\hbar\gamma_{\text{ED}} = 75.7 \mu\text{eV}$, respectively. Consequently, the strong coupling threshold is quantitatively evaluated as $\frac{\hbar\gamma_{\text{qBIC}} + \hbar\gamma_{\text{ED}}}{2} = 54.1 \mu\text{eV}$. Since the Rabi splitting energy ($\hbar\Omega = 79.4 \mu\text{eV}$) strictly exceeds the average damping loss ($54.1 \mu\text{eV}$), the criterion for the strong coupling regime is rigorously satisfied.

APPENDIX B: COMPARATIVE EVALUATION OF SENSING MECHANISMS AND RESONANCE MODES

Table 2 summarizes recent studies on terahertz metasurfaces based on BIC and EIT mechanisms. Notably, our proposed

Table 2. Comparison of Performance with Reported Terahertz Metasurfaces

Reference	Resonance Position	Mechanism	Implementation	Q-factor	Group Delay	Target (LOD)
[8]	0.5–3.0 THz	BIC	Experiment	20	/	cTnl (0.5 pg/mL)
[10]	1.5–2.1 THz	BIC	Experiment	58	/	Hcy (12.5 pmol/μL)
[18]	600–640 nm	BIC + EIT	Experiment	6170	50 μs	/
[19]	0.46–0.52 THz	BIC + EIT	Experiment	10^{13} – 10^{17}	117 ps	/
[46]	2–3.5 μm	BIC + EIT	Simulation	106	GST + switchable	/
This work	0.6–0.9 THz	BIC + EIT	Experiment	102	14.2 ps	Furfural (0.25 ppm)

quasi-BIC system achieves a superior quality factor of 102 and a substantial experimental group delay of 14.2 ps, resulting in a competitive LOD of 0.25 ppm for furfural sensing.

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